Detecting Medicare Provider Fraud Using Supervised and Unsupervised Learning

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Abstract

Healthcare fraud poses a significant challenge to the United States healthcare system, resulting in billions of dollars in annual losses and reducing the efficiency of public insurance programs such as Medicare. This study investigates whether Medicare provider fraud can be effectively identified and predicted using provider-level features derived from inpatient and outpatient claims and beneficiary data. Claim-level information was aggregated into provider-level summary variables representing claim volume, cost patterns, claim complexity, and patient mix. Several machine learning approaches were evaluated, including logistic regression, random forest, XGBoost, and isolation forest. Because fraudulent providers represented a relatively small proportion of the dataset, class imbalance was addressed through weighted modeling approaches and evaluation metrics beyond overall accuracy, including precision, recall, F1-score, and ROC-AUC. Results showed that supervised learning models substantially outperformed an unsupervised isolation forest approach, with random forest achieving the strongest overall balance between fraud precision and recall. Across models, provider claim volume and reimbursement-related variables consistently emerged as the most important predictors of fraud. The findings suggest that relatively interpretable provider-level summary features can support strong fraud classification performance without requiring more computationally expensive deep learning or graph-based methods.

1. Introduction

Health care fraud is the act of intentionally submitting fraudulent claims to health insurers to receive illegal benefits or payments, resulting in billions of dollars of losses in the United States annually [4].

One significant form of health care fraud is **Medicare provider fraud**, in which healthcare providers submit false claims to the Medicare program for reimbursement. Common ways providers engage in this type of fraud include:

- **Double billing**: submitting multiple claims for the same service
- **Phantom billing**: submitting a claim for a service that never occurred
- **Upcoding**: submitting a claim for a more expensive or complex service than the patient actually received

Medicare provider fraud can lead to a multitude of far-reaching effects, including increased insurance premiums, higher taxes, patient harm, and severe legal penalties [4]. Because of these consequences, Medicare fraud detection has become an important area of research in machine learning and healthcare analytics.

The goal of this study is to evaluate whether Medicare provider fraud can be effectively identified and predicted using provider-level features derived from inpatient and outpatient claims and beneficiary data. To do this, individual claim and patient information were aggregated into provider-level summary variables and used as inputs, and multiple machine learning approaches were compared, including logistic regression, random forest, XGBoost, and isolation forest. In addition to measuring predictive performance, this study also examines which provider-level characteristics are most strongly associated with fraud, with the broader aim of identifying a practical and interpretable approach to Medicare fraud detection.

2. Previous Work

Medicare fraud detection has been widely studied using traditional machine learning, deep learning, and more recently, graph-based neural network approaches. Prior work highlights both the potential of predictive models to identify fraud as well as the challenges posed by working with healthcare data, particularly due to class imbalance and feature complexity.

Early work in Medicare fraud detection primarily focused on applying traditional machine learning methods to provider-level data. In [1], the authors conducted a comparative study using supervised, unsupervised, and hybrid machine learning approaches on Medicare Provider Utilization and Payment Data released by the Centers for Medicare and Medicaid Services (CMS). They also leveraged fraud labels obtained from the List of Excluded Individuals/Entities (LEIE) database, which contains identifiers for providers excluded from participation in federal healthcare programs due to violations such as fraud or abuse. Their findings indicate that supervised machine learning methods, such as Random Forest and Gradient Boosted Machine (GBM), generally outperformed unsupervised techniques, particularly when class imbalance was addressed through sampling strategies such as 80–20 undersampling. This study also highlighted the importance of data preprocessing and class distribution in determining model performance, as results varied significantly depending on provider type and sampling method.

Subsequent research has explored the application of deep learning to Medicare fraud detection, particularly when dealing with highly imbalanced datasets. In [3], the authors investigated multiple deep neural network (DNN) approaches using CMS Medicare Part B provider-level data, where fraudulent cases made up as little as 0.03% of observations. Their work emphasized the important role of class imbalance handling and demonstrated that techniques such as random over-sampling (ROS) and hybrid ROS–RUS significantly improved performance, achieving AUC scores above 0.85. This publication established deep learning as a viable approach, but highlighted that performance gains depend heavily on the careful treatment of imbalanced data.

More recent studies have shifted toward incorporating entity relationships through graph-based methods. In [5], the authors propose a framework that models relationships among providers, beneficiaries, and physicians as a graph structure to capture collusion patterns present in fraudulent activity. Their study compared graph neural networks (GNNs) with traditional machine learning models augmented by graph-derived features such as centrality measures. Interestingly, their results showed that conventional machine learning models enhanced with graph centrality features outperformed GNNs in terms of precision, recall, and computational efficiency. In particular, they reported substantial improvements in recall (up to 24%) while requiring significantly less training time than GNN-based approaches. This suggests that while relational modeling is important, simpler feature engineering approaches may be more practical for large-scale fraud detection systems.

Overall, previous work in this space demonstrates a progression from evaluating traditional machine learning model performances to comparisons of deep learning and graph-based approaches. While advanced methods such as GNNs and DNNs can capture more complex patterns in the data, their effectiveness often depends on addressing key challenges such as class imbalance and computational complexity.

Unlike prior work that emphasizes highly complex models or graph-based relational structures, the present study examines whether fraud can be effectively detected using engineered provider-level summary variables and more accessible machine learning methods. By focusing on interpretable predictors derived from claims and beneficiary data, this work highlights a more practical modeling strategy while still achieving strong classification performance. In this way, the study contributes to the literature by showing that useful fraud prediction can still be obtained without relying on computationally expensive deep neural networks or graph-based neural networks.

3. Methodology

3.1. Data

The datasets used for this study were sourced from Kaggle [2] and included 558,211 total Medicare claims (40,474 inpatient + 517,737 outpatient), 138,556 beneficiary records, and a separate provider-level file containing 5,410 provider IDs with associated fraud labels. Because the provided Kaggle dataset description is limited and Medicare data is highly sensitive and subject to HIPAA compliance, the dataset is likely synthetic or simulated by the author.

Exploratory Data Analysis (EDA) was conducted before proceeding with feature engineering and model development. Class imbalance was first assessed. Of the 5,410 total providers in the dataset, 4,904 were non-fraudulent (90.65%) and 506 were fraudulent (9.35%). While class imbalance was certainly present in the data, this reflects the rarity of fraud in real Medicare datasets. Due to this class imbalance, model training used class-weight settings where applicable, and the data was split using stratification so that the fraud proportion remained consistent across both training and validation sets. Additionally, final evaluations focused not only on accuracy but on other significant metrics such as precision, recall, F1-score, and ROC-AUC.

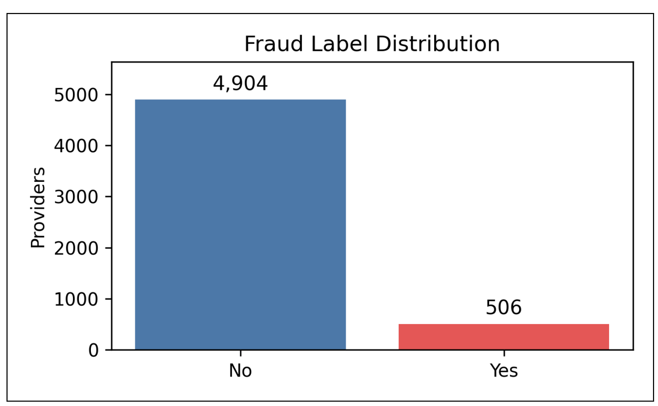

Figure 1. Fraud Class Distribution

Missingness was also evaluated, but was ultimately not a major issue for the claim-level datasets because most of the missing values came from fields where a blank value was informative rather than problematic. For example, missing physician or procedure code fields meant that a claim did not involve that specific role or did not require that many codes. Since the final models ultimately used provider-level aggregated features rather than relying on individual fields, this missingness had limited impact on the analysis, though two fields (ClmProcedureCode_6 from the inpatient dataset and ClmProcedureCode_5 and ClmProcedureCode_6 from the outpatient dataset) were removed due to 100% missingness. Missing deductible amounts were filled in with 0s across both claim datasets.

Finally, claim reimbursement amount and claim length were specifically explored to compare payment patterns and claim duration across both inpatient and outpatient claims. This analysis showed that inpatient claims generally had much higher reimbursement amounts and longer claim lengths on average, while outpatient claims were concentrated at lower amounts. This makes sense, as patients staying for multiple days in a hospital would generally require more intensive care and therefore generate higher reimbursement amounts than patients receiving outpatient services. However, these differences suggest that the two claim types operate on significantly different cost and time scales, which provided useful context for later feature engineering and model development.

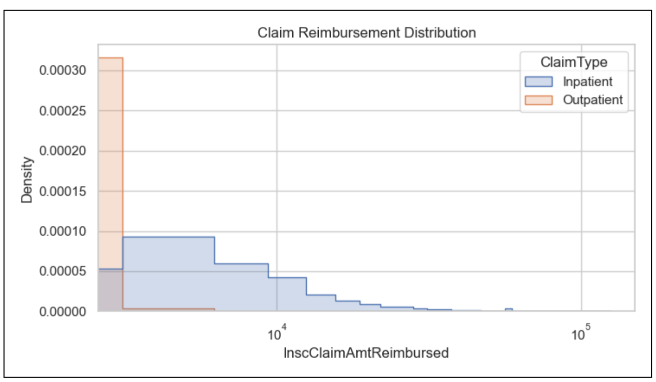

Figure 2. Claim Reimbursement Distribution

3.2. Feature Engineering

The feature engineering process focused on transforming the raw inpatient, outpatient, and beneficiary datasets into provider-level summary variables that could act as inputs for the fraud classification models. For the inpatient and outpatient claim-level data, date fields were converted to a datetime format so a new ClaimLength variable could be calculated as the number of days between claim start and end dates. The diagnosis code fields were counted

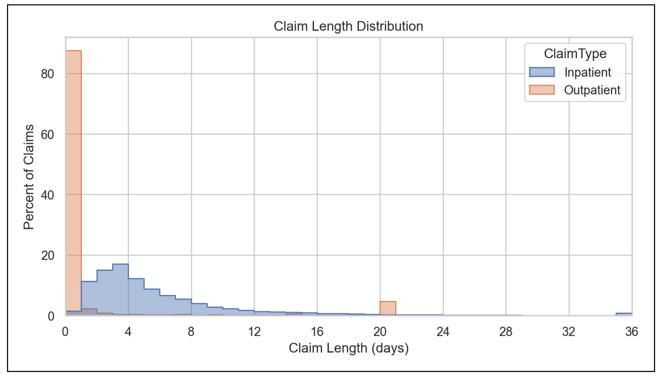

Figure 3. Claim Length Distribution

and summarized into the variable NumDiagnoses, representing the number of diagnoses recorded for each claim, and a HasAdmitDiagnosis indicator variable was created to capture whether an initial admission diagnosis was present. From the beneficiary dataset, DOB was used to derive a new Age variable, the chronic condition fields were counted and combined into the NumChronicConditions variable, and RenalDiseaseIndicator was converted into a binary RenalDisease feature. These beneficiary-level features were then merged into the inpatient and outpatient claims data using beneficiary ID.

After feature engineering, the claim and beneficiary information were aggregated to the provider level because the prediction task was to classify providers (rather than individual claims) as fraudulent or non-fraudulent. The inpatient and outpatient summaries were combined by provider ID and then joined with the labeled provider file containing the fraud labels. This produced the final modeling dataset.

After experimenting with different combinations of features, the final model used 20 provider-level input variables. These variables fall under four main groups: Claim Volume, Cost Patterns, Claim Complexity, and Patient Mix. Table 1 summarizes the feature groups and the engineered features included in each.

4. Results

4.1. Logistic Regression

Logistic regression was used as a baseline model because it provided a simple and interpretable starting point for binary fraud classification. The provider-level features were split into training and validation sets using an 80-20 stratified split so that the fraud proportion remained consistent across both sets. Because logistic regression is sensitive to feature scale, the predictors were also standardized before training. To address the previously stated class imbalance, the model used ”balanced” class weighting, which placed greater emphasis on the minority fraud class during train-

Feature Group	Features Used
Claim Volume	Claim count
Cost Patterns	Average reimbursement amount
	SD reimbursement amount
	Average deductible amount
Claim Complexity	Average # of diagnoses
	Average claim length
	Share with admission diagnosis
Patient Mix	Average age
	Average # of chronic conditions
	Renal disease rate

Table 1. Summary of feature groups and engineered features

ing. Model performance was evaluated using classification metrics and ROC-AUC, and several probability thresholds were tested on the validation set. Model coefficients were also examined to identify which predictors were most strongly associated with fraud classification. A threshold of 0.7 produced the strongest overall performance, suggesting that a stricter cutoff improved the precision-recall tradeoff for identifying fraudulent providers. On the validation set, this threshold produced an accuracy of 0.918, fraud-class precision of 0.541, recall of 0.792, and an F1-score of 0.643, with a ROC AUC of 0.953. The largest coefficients were observed for inpatient claim count (1.590), outpatient claim count (1.195), mean inpatient deductible amount paid (0.521), mean inpatient claim length (0.461), standard deviation of inpatient reimbursement amount (0.447).

4.2. Random Forest

Random forest was then applied as a nonlinear ensemble alternative to logistic regression. This model builds many decision trees and combines their predictions, allowing it to capture more complex relationships and interactions among the provider-level features without requiring feature scaling. The model was trained with 300 trees and class-weight balancing to account for the relatively small number of fraudulent providers (n=405 after the train/validation split). Performance was evaluated using classification metrics and ROC-AUC. On the validation set, random forest achieved an accuracy of 0.941, fraud-class precision of 0.713, recall of 0.614, and an F1-score of 0.660, with a ROC AUC of 0.956. Feature importance scores showed that the most influential variables included inpatient claim count (0.145), standard deviation of inpatient reimbursement amount (0.107), outpatient claim count (0.104), mean inpatient reimbursement amount (0.091), mean inpatient claim length (0.088).

4.3. XGBoost

XGBoost was used as a more advanced gradient boosting-based classification model, designed to improve predictive performance by building trees sequentially and correcting prior errors at each stage. The model was configured with 300 estimators, a moderate tree depth (4), a small learning rate (0.05), and subsampling of both rows and features to improve generalization. Because the data was imbalanced, scale_pos_weight was set based on the ratio of non-fraudulent to fraudulent providers in the training data so that the model would place more emphasis on detecting the minority class. As with the other supervised models, performance was evaluated using classification metrics and ROC-AUC on the validation set, and feature importance scores were examined to better understand which variables were most influential. On the validation set, XGBoost produced an accuracy of 0.921, fraud-class precision of 0.553, recall of 0.772, and an F1-score of 0.645, with a ROC AUC of 0.953. Feature importance values indicated that inpatient claim count was again the strongest predictor (0.207), followed by standard deviation of inpatient reimbursement amount (0.156), outpatient claim count (0.097), mean inpatient claim length (0.068), mean inpatient reimbursement amount (0.053).

4.4. Isolation Forest

Isolation forest was included as an unsupervised anomaly-detection approach. Rather than learning directly from both fraud and non-fraud labels, this method was trained only on non-fraudulent providers so that unusual provider behavior could be identified as potential anomalies. The contamination parameter was set to approximately match the observed fraud rate in the training data (about 9.4%) which allowed the model to flag a similar proportion of cases as anomalous. Predictions from the isolation forest were then converted into binary fraud flags, and anomaly scores were used to compute ROC-AUC on the validation set. This approach was useful because healthcare fraud can be viewed as a rare and abnormal pattern, making anomaly detection a reasonable alternative to fully supervised classification. Unlike logistic regression, random forest, and XGBoost, the isolation forest model did not provide a straightforward feature importance or coefficient-based interpretation in this analysis. Its validation performance was also substantially weaker than the supervised models, with an accuracy of 0.834, fraud-class precision of 0.224, recall of 0.317, an F1-score of 0.262, and a ROC AUC of 0.808.

Note: Training metrics and validation metrics are included in Tables 2 and 3 for reference

Table 2. Training Performance by Model

Model	Acc.	Fraud Prec.	Fraud Recall	Fraud F1	ROC AUC
Logistic Reg. ($t = 0.70$)	0.909	0.509	0.709	0.592	0.931
Random Forest	0.997	0.969	1.000	0.984	1.000
XGBoost	0.956	0.680	0.998	0.809	0.997
Isolation Forest	0.848	0.237	0.281	0.257	0.794

Table 3. Validation Performance by Model

Model	Acc.	Fraud Prec.	Fraud Recall	Fraud F1	ROC AUC
Logistic Reg. ($t = 0.70$)	0.918	0.541	0.792	0.643	0.953
Random Forest	0.941	0.713	0.614	0.660	0.956
XGBoost	0.921	0.553	0.772	0.645	0.953
Isolation Forest	0.834	0.224	0.317	0.262	0.808

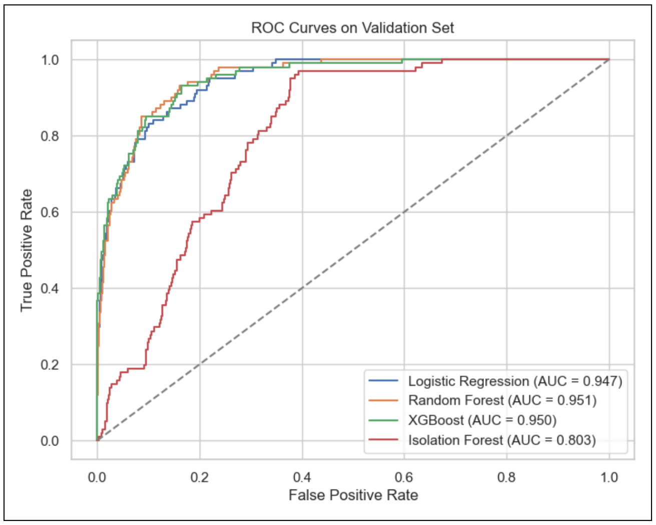

Figure 4. Validation Set ROC Curves

5. Discussion

The results suggest that provider-level fraud prediction is best addressed as a supervised classification problem rather than as an unsupervised anomaly-detection task. Across the supervised models, performance was consistently strong, with logistic regression, random forest, and XGBoost all achieving ROC-AUC values near 0.95 on the validation set. This indicates that the engineered provider-level features contained meaningful signal for distinguishing fraudulent from non-fraudulent providers. In contrast, isolation forest performed substantially worse across all major metrics, particularly on fraud-class precision, recall, and F1-score. Although fraud can conceptually be viewed as “anomalous” behavior, these results suggest that fraud cases in this dataset were better captured through direct supervised learning than through general anomaly detection.

Among the supervised approaches, the tradeoff between fraud precision and fraud recall is especially important. Logistic regression and XGBoost achieved the strongest fraud recall, identifying a larger share of fraudulent providers than random forest. This is valuable in fraud detection contexts, where failing to identify fraudulent behavior may be more costly than flagging additional non-fraudulent providers for review . However, both models did so at the cost of lower fraud precision, meaning that a greater proportion of flagged providers were false positives. Random forest produced the highest fraud precision and the strongest fraud-class F1-score, indicating a more balanced performance between identifying fraud and limiting false alarms. Taken together, these results suggest that random forest offered the best overall balance in this study, while logistic regression and XGBoost may still be attractive when maximizing fraud recall is the higher priority.

The feature importance and coefficient patterns also provide insight into the types of provider behavior associated with fraud prediction. Across models, claim volume was consistently one of the strongest signals, as both inpatient and outpatient claim counts appeared among the most important predictors. Measures related to reimbursement behavior, particularly the mean and standard deviation of reimbursement amounts, were also important. In addition, inpatient claim length and patient-mix variables, such as the average number of chronic conditions, contributed to prediction. These findings suggest that fraudulent providers may differ not only in how much they bill, but also in the volume, variability, and clinical characteristics of the claims associated with them. At the same time, these variables should not be interpreted causally; they indicate statistical association with the fraud labels rather than ”proof” of fraudulent behavior.

Several limitations should be considered when interpreting these findings. First, the analysis relied on provider-level aggregation, which was appropriate for the prediction target but may have obscured important claim-level patterns. Secondly, the class imbalance, while not extreme, still affected the relationship between precision and recall and required the use of class weighting and threshold adjustment. Finally, since the dataset was obtained from Kaggle and contained limited documentation, the similarity of this dataset to real Medicare datasets is not fully clear. It is possible model performance in this context may not directly translate to real operational Medicare fraud detection settings.

Overall, the findings indicate that relatively straightforward provider-level feature engineering can support strong fraud classification performance. Even a simple baseline logistic regression performed well, while random forest and XGBoost provided modest additional gains and highlighted the importance of claim count and reimbursement-related variables.

6. Conclusion

This study demonstrates that Medicare provider fraud can be effectively predicted using engineered provider-level features and relatively accessible machine learning techniques. Across all supervised models tested, strong classification performance was achieved, with ROC-AUC values near 0.95 on the validation set. Among the evaluated approaches, random forest produced the best overall balance between fraud precision and recall, while logistic regression and XGBoost achieved slightly stronger fraud recall. In contrast, the unsupervised isolation forest model performed substantially worse, suggesting that supervised learning methods are better suited for identifying fraudulent provider behavior within this dataset.

The results also highlight the importance of provider claim volume and reimbursement-related variables in fraud prediction. Features such as inpatient and outpatient claim counts, reimbursement variability, and measures of claim complexity consistently appeared among the strongest predictors across models. These findings suggest that fraudulent providers may exhibit distinguishable billing and utilization patterns that can be captured through provider-level aggregation and feature engineering.

This study demonstrates that straightforward feature engineering combined with supervised machine learning can provide a practical and interpretable framework for Medicare fraud detection. Future work could extend this analysis by testing additional validation strategies and evaluating whether claim-level or even hybrid provider-claim approaches can improve fraud detection further.

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